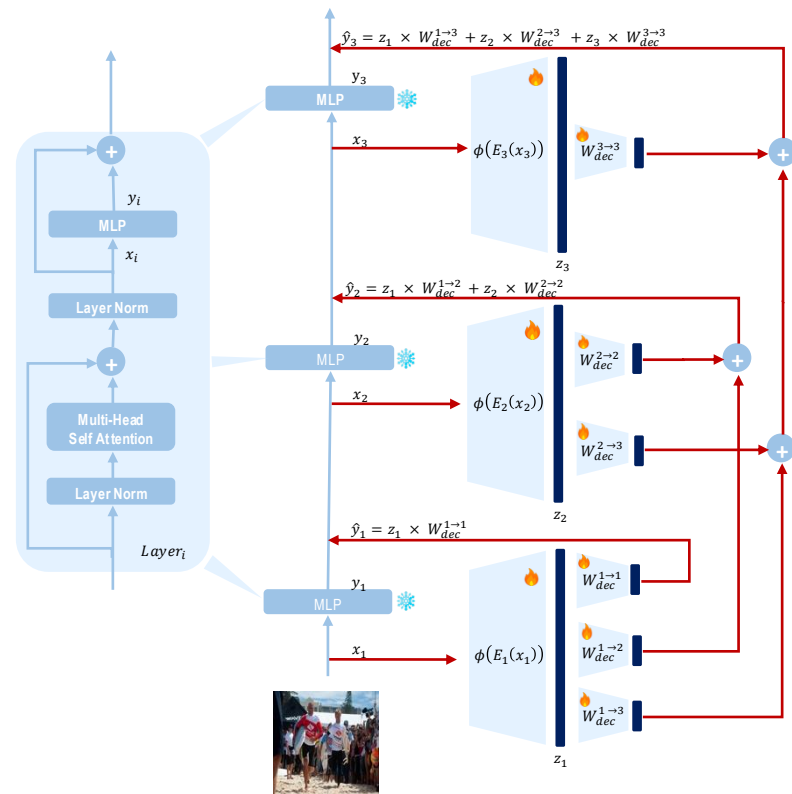


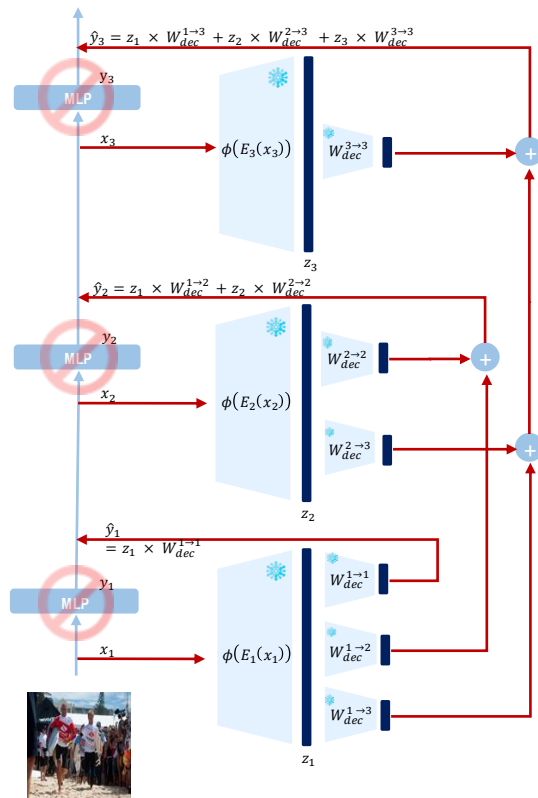
Training: CLTs

Use pre – MLP activations
to predict target layer's post – MLP activation



Inference: Replacement Model

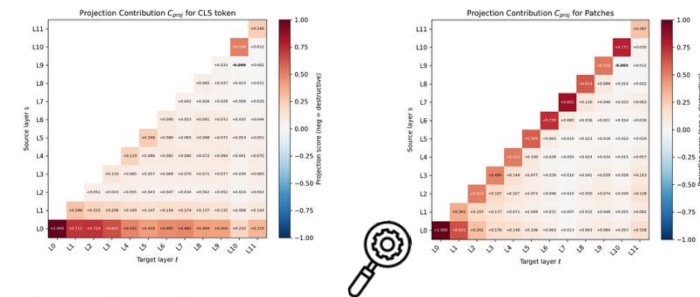
Replace MLP blocks with learned CLTs



ViT Understanding through CLTs

Use CLTs as an interpretable proxy for ViTs

$$C_{i \rightarrow j}^{proj} = E_x \left[\left(\frac{1}{T} \sum_{t=1}^T \frac{\langle c_{i \rightarrow j}^{(t)}, \hat{y}_j^{(t)}(x) \rangle}{\|\hat{y}_j^{(t)}\|_2^2} \right) \right]$$



Is there a structural difference in cross – layer contributions between CLS and patches ?

Which layers are more significant in shaping the final representation?